

Q6 For circular motion at the highest point,

$$mg - N = \frac{mv^2}{r}$$

$$W - \frac{W}{3} = \frac{mv^2}{r}$$

$$mg - \frac{mg}{3} = \frac{mv^2}{r}$$

$$\frac{2mg}{3} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{2gr}{3}}$$

Answer: B

Q7 $F = N = F$